
A Korean Benchmark for Safe Exercise Prescription: From Exam Knowledge to Clinical Action in Metabolic Syndrome and Sarcopenia

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Abstract

Large language models (LLMs) demonstrate near-expert performance on medical licensing-style examinations including Korean health certification tests, yet their capacity for safe clinical deployment remains uncertain. Exercise prescription demands rigorous numerical precision and safety screening, where errors in intensity calculations or risk/medication screening can endanger vulnerable populations. We develop Korean benchmark evaluating LLMs on both theoretical knowledge and clinical exercise prescription for metabolic syndrome and sarcopenia. We assessed six models (frontier: GPT 5.2 Pro, Claude 4.5 Opus, Gemini 3.0 Pro, and open-weight: Gemma 3: 4B/12B/27B) across 500 Korean-language Health Exercise Manager certification exam questions (2020–2024) and 120 clinical vignettes derived from the Korea National Health and Nutrition Examination Survey. Using American College of Sports Medicine (ACSM) guideline-based automated rubrics validated against human experts (ICC=0.761; κ =0.760), we evaluated diagnostic accuracy, intensity calculations, safety protocols, and prescription quality. Frontier models achieved 84.9–95.5% exam accuracy and 95.7–98.3% rubric scores on exercise prescription, while open-weight models scored 40.1–62.5% and 63.1–74.6%, respectively. All models demonstrated reduced accuracy on multimodal questions requiring visual processing compared to text-only questions. Critically, open-weight models exhibited severe failures in intensity calculations (6.4–22.6%) despite moderate exam performance (40.1–62.5%), demonstrating striking bidirectional knowledge-application dissociation where passive recognition fails to guarantee active procedural execution. Our findings suggest that exam-style accuracy is an insufficient proxy for safety-critical procedural reliability, especially for multi-step numerical tasks that warrant human oversight.

1 Introduction

Large Language Models (LLMs) have recently demonstrated near-expert performance on professional medical examinations, prompting clinical deployment considerations as clinical decision support tools [1]. In the general medical domain, specialized models such as Med-PaLM and Med-PaLM 2 have reported exceptional results on standardized benchmarks [2]. Beyond general medicine, domain-tuned models like PH-LLM have shown competitive proficiency in specialized areas, including sleep and fitness coaching certification exams [3].

However, growing evidence suggests that exam-style accuracy is a necessary but insufficient proxy for real-world clinical utility. Standardized licensing examinations fail to capture the uncertainty, safety constraints, and longitudinal reasoning inherent in actual practice [4]. Systematic reviews further reveal that current evaluations rarely utilize real patient-care data, leaving critical dimensions including safety and bias remain under-assessed [5]. Crucially, randomized trials demonstrate that high benchmark proficiency does not reliably translate to improved clinical outcomes or decision-making in real-world workflows [6, 7].

This gap is particularly consequential in exercise prescription, which functions not as generic advice but as a structured “dose” defined by the FITT principle (Frequency, Intensity, Time, Type). Such prescriptions demand rigorous numerical precision and safety screening, where errors in intensity calculation or failure to account for comorbidities can directly endanger vulnerable patients [8]. This challenge is acute in Korea, where metabolic syndrome affects roughly 30% of adults [9, 10], and sarcopenia prevalence rises steeply with age [11].

To our knowledge, no existing benchmark evaluates both theoretical mastery and clinical safety in exercise prescription [12]. Moreover, the prevalence of dataset contamination necessitates rigorous design choices to mitigate memorization effects and ensure true robustness [13, 14]. Broader evaluation frameworks further emphasize that such methodological rigor is essential for distinguishing genuine reasoning from rote recall [15]. Furthermore, systematic reasoning flaws such as hallucinations in numerical interpretation remain persistent failure modes that ‘expert-level’ surface fluency can easily mask [16–18].

Our contributions are threefold: (i) we curate a Korean evaluation suite combining 500 Health Exercise Manager exam questions (2020–2024) with 120 clinical vignettes derived from the Korea National Health and Nutrition Examination Survey (KNHANES) representing patients with metabolic syndrome and sarcopenia; (ii) we propose an 18-item American College of Sports Medicine (ACSM) guideline-based [8] binary rubric for clinical exercise prescriptions and validate an automated LLM-as-a-judge [19, 20] scoring pipeline against an exercise physiologist; and (iii) we quantify modality gaps and knowledge-to-action dissociations across exam-style questions and vignette-based prescriptions.

2 Methods

2.1 Datasets

2.1.1 Health Exercise Manager Certification Exam Dataset

We obtained publicly available examination materials from the Korea Sports Promotion Foundation (KSPO) official examination repository (<https://sqms.kspo.or.kr/exam/wrOnaList.kspo>). The Health Exercise Manager certification consists of eight subject areas; we selected five domains directly relevant to exercise prescription for patients with metabolic syndrome and sarcopenia: exercise physiology, physical fitness assessment, exercise prescription, exercise stress testing, and clinical exercise physiology. Each annual examination allocates 20 questions per domain. This subset comprises five consecutive examination years (2020–2024), yielding 500 multiple-choice questions (20 questions $\times$ 5 domains $\times$ 5 years) with four answer options each. The detailed distribution of questions by domain and year is provided in Supplementary Table 1.

Sixty-seven questions (13.4%) incorporate visual elements requiring multimodal processing, including physiological diagrams, ECG tracings, anatomical illustrations, and questions with complex tabular answer options not amenable to plain-text representation. Raw PDF files were converted to structured JSON format, with embedded images extracted as separate files. All materials underwent human validation to ensure parsing accuracy. The finalized dataset structure includes question text, answer options, correct answer index, subject domain, and image file paths where applicable. A representative sample question is shown in Supplementary Figure 1.

2.1.2 KNHANES-based Clinical Vignette Dataset

KNHANES dataset. We constructed clinical vignettes from KNHANES, a nationally representative cross-sectional survey implemented by the Korea Disease Control and Prevention Agency (KDCA). The KNHANES framework comprises three distinct investigative domains: health interviews, physical examinations, and nutritional assessments [21]. All data collection procedures were conducted in strict adherence to the Declaration of Helsinki and the Bioethics and Safety Act of the Republic of Korea, with written informed consent obtained from every participant. The study protocols for the 2022–2024 KNHANES were reviewed and approved by the Institutional Review Board (IRB) of the KDCA (Approval Nos. 2018-01-03-4C-A, 2022-11-16-R-A, and 2022-11-16-R-03). KNHANES data are publicly available at <https://knhanes.kdca.go.kr/knhanes/main.do>.

Data Preprocessing and Eligibility Criteria. The raw dataset from KNHANES 2022–2024 encompassed 20,191 participants with 26 anthropometric, physiological, and biochemical variables relevant to metabolic syndrome and sarcopenia. Participants with missing values in essential diagnostic variables were excluded from the analysis. These variables encompassed anthropometric measurements (height, weight, waist circumference, body fat mass), segmental muscle mass via bioelectrical impedance analysis, grip strength, cardiovascular parameters (resting heart rate, blood pressure), metabolic biomarkers (fasting glucose, triglycerides, HDL cholesterol), and medication history. Consequently, a total of 9,780 individuals with complete records were retained.

Diagnostic Classification. Metabolic syndrome was diagnosed based on the modified National Cholesterol Education Program Adult Treatment Panel III (NCEP-ATP III) criteria [22], applying the specific abdominal obesity cutoffs for the Korean population [23]. Participants were classified as having metabolic syndrome if they met three or more of the following five conditions: (1) Abdominal Obesity: waist circumference ≥ 90 cm for men or ≥ 85 cm for women; (2) Hypertriglyceridemia: serum triglycerides ≥ 150 mg/dL or current use of lipid-lowering medication; (3) Reduced HDL-Cholesterol: < 40 mg/dL for men and < 50 mg/dL for women, or specific pharmacological treatment for dyslipidemia; (4) Hypertension: systolic blood pressure ≥ 130 mmHg, diastolic blood pressure ≥ 85 mmHg, or current use of antihypertensive therapy; and (5) Hyperglycemia: fasting plasma glucose ≥ 100 mg/dL or current use of insulin or oral hypoglycemic agents. Sarcopenia was identified per Asian Working Group for Sarcopenia (AWGS) 2019 guidelines requiring both low appendicular skeletal muscle mass index (SMI: $\text{ASM}/\text{height}^2 < 7.0$ kg/m² in men, < 5.7 kg/m² in women) and low grip strength (< 28 kg in men, < 18 kg in women) [24].

Composition-Matched Sampling. We constructed two non-overlapping vignette sets: a metabolic syndrome set ($n=60$) and a sarcopenia set ($n=60$), totaling 120 vignettes. For each set, we performed stratified sampling by sex (male/female), three age groups (20–39, 40–64, and ≥ 65 years), and the corresponding disease status (present/absent), yielding 12 strata per set.

Vignette Generation. Each vignette presented patient information in a standardized Korean-language format comprising five sections: (1) demographic profile (age, sex), (2) anthropometric and body composition data (height, weight, waist circumference, body fat mass, appendicular skeletal muscle mass), (3) functional capacity (maximum grip strength from four trials), (4) cardiovascular and metabolic parameters (resting heart rate, blood pressure, fasting glucose, triglycerides, HDL cholesterol), and (5) current medication regimen. Ground truth labels included binary diagnostic outcomes (metabolic syndrome, sarcopenia), individual NCEP-ATP III component flags, and calculated skeletal muscle index for automated evaluation. A representative clinical vignette example is provided in Supplementary Figure 2.

2.2 Large Language Model Configuration

2.2.1 Model Selection

We evaluated six large language models spanning three capability tiers to assess the relationship between model scale and clinical reasoning performance. Frontier models included GPT 5.2 Pro [25] (OpenAI, accessed January 2026), Claude 4.5 Opus [26] (Anthropic, accessed January 2026), and Gemini 3.0 Pro [27] (Google, accessed January 2026), accessed via their respective application programming interface (API) endpoints with default parameters. Open-weight models comprised the Gemma 3 series [28] at three parameter scales (4B, 12B, and 27B), deployed locally via Ollama v0.14.0 [29] to enable reproducible inference without external API dependencies. All models were evaluated using their default parameters as provided by the respective platforms, ensuring alignment with typical deployment scenarios.

2.2.2 Prompt for Health Exercise Manager Exam

For the certification examination dataset, we employed a role-based prompting strategy that positioned models as ACSM Certified Clinical Exercise Physiologists [30], allied health professionals trained to utilize prescribed exercise and health behavior interventions for individuals with chronic diseases. The prompt structure comprised three components: (1) role assignment specifying clinical expertise, (2) task instruction to select the correct answer from four options, and (3) output format constraint requiring only the answer number. For multimodal questions containing visual elements, images were integrated into prompts using platform-specific formats: base64-encoded strings for Claude 4.5 Opus and Gemma 3 models, file identifiers for GPT 5.2 Pro, and

image bytes for Gemini 3.0 Pro. When supplementary context was present, it was inserted between the question text and answer choices to maintain logical flow. The complete prompt template structure is provided in Supplementary Figure 3.

2.2.3 Prompt for Clinical Vignette Dataset

For the clinical vignette dataset, models were instructed to function as clinical exercise physiologists generating ACSM guideline-based exercise prescriptions. The prompt mandated four output sections: (1) risk assessment for metabolic syndrome (NCEP-ATP III) and sarcopenia (AWGS 2019), (2) target heart rate calculation via Karvonen formula [31], (3) safety precautions for identified conditions and medications, and (4) FITT-principle exercise prescription specifying frequency, intensity, time, and type. To facilitate automated evaluation, responses were constrained to structured Markdown format without preambles or postambles. All vignette prompts were text-only. The complete prompt template is provided in Supplementary Figure 4.

2.3 Evaluation Framework

2.3.1 Health Exercise Manager Certification Exam Evaluation

All models were evaluated using a three-run protocol to assess response stability. For each model and trial, responses were generated independently with default sampling parameters. Accuracy was calculated as the proportion of correct answers across the 500-question dataset, with support for multiple valid answers where specified by exam guidelines. Performance metrics were computed overall and stratified by five subject domains. Results are reported as mean±standard deviation (SD) across the three independent runs. To assess temporal robustness, we analyzed year-over-year performance trends from 2020 to 2024, calculating accuracy separately for each exam year. Additionally, we quantified the modality gap by comparing accuracy on text-only questions versus multimodal questions requiring image interpretation.

2.3.2 Clinical Vignette Dataset Evaluation

Rubric Development. To evaluate exercise prescriptions generated for the clinical vignette dataset, we developed an ACSM guideline-based structured rubric assessing 18 binary criteria across four domains: (1) diagnostic accuracy for metabolic syndrome risk factors and sarcopenia (7 items), (2) exercise intensity calculation precision using the Karvonen formula (3 items), (3) personalized safety management for medications and age-related risks (3 items), and (4) FITT prescription quality and guideline adherence (5 items). Each criterion was scored as correct (1) or incorrect (0), yielding a maximum total score of 18 points. The complete rubric with detailed scoring criteria is provided in Supplementary Table 2.

Automated Evaluation System. To ensure scalable and consistent evaluation, we employed the highest-performing model from the certification exam as an automated evaluator. The model was prompted to assess each prescription against the 18-item rubric following established LLM-as-a-judge methodologies [32]. The evaluation prompt instructed the model to first establish ground truth by independently calculating diagnostic criteria per NCEP-ATP III and AWGS 2019 guidelines, then compare model-generated prescriptions against this reference using strict binary scoring rules. For target heart rate calculations, responses within ±5 bpm of the correct Karvonen-derived range were considered accurate. The complete evaluation prompt template is provided in Supplementary Figure 5.

Reliability Validation. To validate automated scoring reliability, an exercise physiologist with a master's degree in exercise physiology independently evaluated 100 randomly sampled vignettes using the identical rubric provided to the automated evaluator. Inter-rater agreement was quantified using intraclass correlation coefficient (ICC 2,1) [33], quadratic weighted Cohen's kappa [34], Pearson correlation coefficient, Spearman rank correlation coefficient, Kendall rank correlation coefficient (τ) [35], and mean absolute error [36].

Performance Metrics and Gap Analysis. For the full 120-vignette dataset, all model-generated prescriptions across three runs were evaluated using the validated automated scoring system. Scores were aggregated by the four clinical domains and reported as mean±SD across three independent runs. To examine the knowledge-application gap, we calculated the difference between overall exam accuracy and normalized automated evaluation scores (total score/18 × 100%).

2.4 Statistical Analysis

All statistical analyses were performed using Python (v3.11.5) with SciPy (v1.11.1) and scikit-learn (v1.4.0) libraries. Descriptive statistics for performances of LLMs are reported as mean $\pm$ SD across three independent runs. Statistical significance for correlation coefficients was defined as $p < 0.05$, with $p < 0.001$ denoted by ***.

3. Results

3.1 Performance on Health Exercise Manager Certification Exam

Overall Performance. Overall performance on the 500-question Health Exercise Manager certification exam is presented in Table 1. Accuracy ranged from 40.1% (Gemma 3:4b) to 95.5% (Gemini 3.0 Pro), with frontier models outperforming smaller open-weight variants. Gemini 3.0 Pro achieved the highest overall accuracy (95.5 $\pm$ 0.5%), followed by GPT 5.2 Pro (92.8 $\pm$ 0.7%) and Claude 4.5 Opus (84.9 $\pm$ 0.2%). Among open-weight models, performance scaled with model size: Gemma 3:27b (62.5 $\pm$ 0.6%) outperformed Gemma 3:12b (55.8 $\pm$ 0.5%) and Gemma 3:4b (40.1 $\pm$ 0.6%).

Table 1: Comprehensive performance analysis on the Health Exercise Manager exam

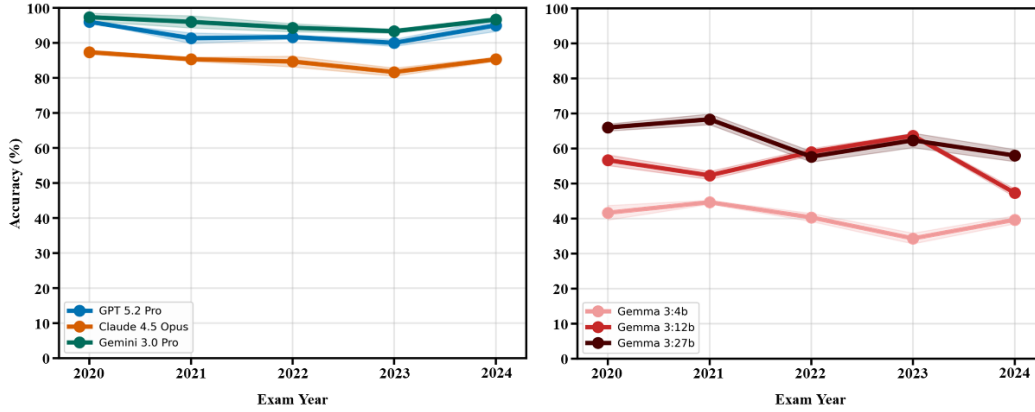
Accuracy (%, mean $\pm$ SD)	GPT 5.2 Pro	Claude 4.5 Opus	Gemini 3.0 Pro	Gemma 3:4b	Gemma 3:12b	Gemma 3:27b
Overall	92.8 $\pm$ 0.7	84.9 $\pm$ 0.2	95.5 $\pm$ 0.5	40.1 $\pm$ 0.6	55.8 $\pm$ 0.5	62.5 $\pm$ 0.6
Exercise physiology	94.7 $\pm$ 0.6	88.3 $\pm$ 0.6	95.0 $\pm$ 1.0	33.7 $\pm$ 0.6	60.3 $\pm$ 0.6	69.0 $\pm$ 0.0
Physical fitness assessment	93.0 $\pm$ 1.7	79.7 $\pm$ 1.5	94.3 $\pm$ 1.2	39.7 $\pm$ 0.6	51.7 $\pm$ 1.5	57.7 $\pm$ 2.3
Exercise prescription	86.0 $\pm$ 1.0	82.0 $\pm$ 1.0	92.7 $\pm$ 2.1	42.0 $\pm$ 1.7	48.7 $\pm$ 1.5	55.7 $\pm$ 1.5
Exercise stress testing	90.3 $\pm$ 1.2	77.7 $\pm$ 2.5	95.7 $\pm$ 1.2	35.3 $\pm$ 0.6	44.0 $\pm$ 2.0	47.0 $\pm$ 1.7
Clinical exercise physiology	100.0 $\pm$ 0.0	96.7 $\pm$ 0.6	100.0 $\pm$ 0.0	50.0 $\pm$ 1.0	74.3 $\pm$ 0.6	83.0 $\pm$ 0.0

Domain-specific Performance. Domain-specific performance revealed heterogeneity in task difficulty. Clinical exercise physiology questions proved most accessible, with Gemini 3.0 Pro and GPT 5.2 Pro both achieving perfect accuracy (100.0 $\pm$ 0.0%), and even the smallest model (Gemma 3:4b) reaching 50.0 $\pm$ 1.0%. In contrast, exercise stress testing posed greater challenges, with frontier models scoring 77.7 to 95.7% and Gemma variants dropping to 35.3 to 47.0%. The exercise prescription domain, directly relevant to clinical vignette tasks, showed intermediate difficulty with accuracy ranging from 42.0 to 92.7% across models. Notably, all models demonstrated low run-to-run variability (SD $<$ 2.6% across all conditions), indicating stable performance across independent evaluations.

Temporal Robustness. Temporal robustness and modality effects are shown in Figure 1. Temporal analysis revealed performance across examination years, with frontier models exhibiting year-to-year fluctuations within 10 percentage points from 2020 to 2024 (Figure 1a). GPT 5.2 Pro and Gemini 3.0 Pro maintained above 90% accuracy across all years, while Claude 4.5 Opus consistently scored above 80%. Open-weight models displayed greater variability, with Gemma variants showing fluctuations ranging from 10 to 20 percentage points across the five-year period.

Modality Gap. The modality gap analysis demonstrated that multimodal questions requiring visual processing were more difficult than text-only counterparts (Figure 1b). All models exhibited accuracy decreases on image-based questions, with Gemini 3.0 Pro showing the smallest performance gap, followed by GPT 5.2 Pro and Claude 4.5 Opus. Among open-weight models, smaller variants (Gemma 3:4b and 12b) demonstrated relatively smaller modality gaps compared to the larger Gemma 3:27b model.

(a) Temporal Robustness



(b) Modality Gap

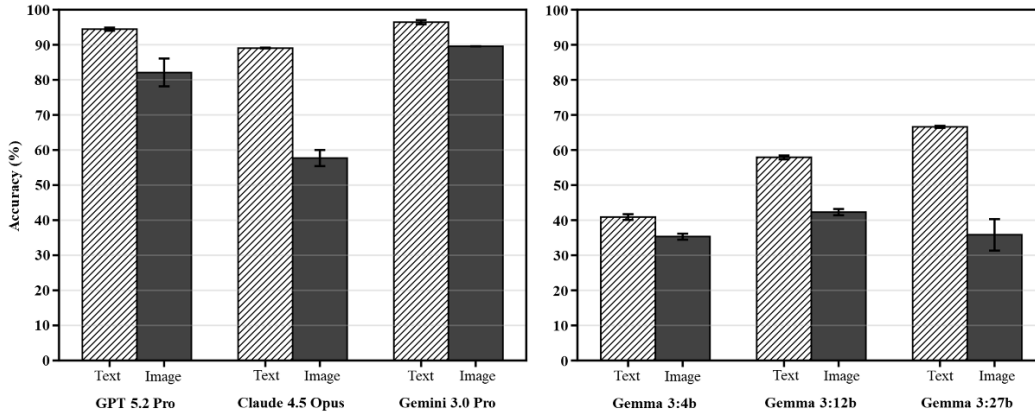


Figure 1: Performance consistency and modality analysis

3.2 Validation of Automated Evaluation System

Inter-Rater Reliability Between Automated Judge and Human Expert. We compared scores from the LLM-as-a-judge evaluator with those from an exercise physiologist on 100 randomly sampled clinical vignettes. Inter-rater agreement metrics are presented in Table 2. Association measures were as follows: Pearson correlation ($r = 0.846$, $p < 0.001$), Spearman rank correlation ($\rho = 0.828$, $p < 0.001$), and Kendall rank correlation ($\tau = 0.706$, $p < 0.001$). Absolute agreement metrics yielded intraclass correlation coefficient (ICC 2,1) of 0.761 and quadratic weighted Cohen's kappa of 0.760. The mean absolute error was 1.875 points on the 18-point scale.

Table 2: Agreement between the Automated Judge and Human Expert Consensus

Category	Evaluation Metric	Value
Association Measures	Pearson Correlation (r)	0.846***
	Spearman Rank Correlation (ρ)	0.828***
	Kendall Rank Correlation (τ)	0.706***
Inter-rater Agreement	Intraclass Correlation (ICC (2,1))	0.761
	Quadratic Weighted Kappa (κ)	0.760
Error Magnitude	Mean Absolute Error (MAE)	1.875

Item-Level Agreement Analysis. Item-level agreement rates between the automated judge and human expert across the 18 rubric criteria are shown in Figure 2. All items achieved above 80% agreement except for two criteria: metabolic syndrome diagnosis (Q6: 63.3%) and internal consistency of heart rate ranges (Q17: 58.3%).

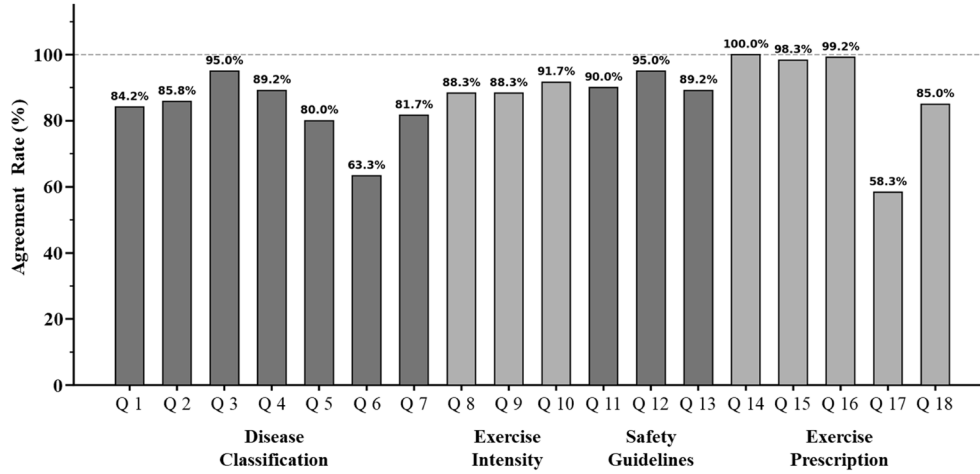


Figure 2: Item-level agreement rates between automated judge and human expert

3.3 Performance on Clinical Vignette Exercise Prescriptions

Overall Performance. Model performance on the 120 clinical vignette dataset is presented in Table 3. Overall accuracy ranged from 63.1% (Gemma 3:4b) to 98.3% (Gemini 3.0 Pro). Frontier models achieved Gemini 3.0 Pro (98.3±4.5%), GPT5.2 Pro (98.0±4.5%), and Claude 4.5 Opus (95.7±5.9%). Open-weight models scored Gemma 3:12b (74.6±10.4%), Gemma 3:27b (71.8±8.3%), and Gemma 3:4b (63.1±10.4%).

Table 3: Comprehensive performance analysis on the exercise prescriptions

Binary rubric accuracy (% , mean±SD)	GPT 5.2 Pro	Claude 4.5 Opus	Gemini 3.0 Pro	Gemma 3:4b	Gemma 3:12b	Gemma 3:27b
Overall	98.0 ±4.5	95.7 ±5.9	98.3 ±4.5	63.1 ±10.4	74.6 ±10.4	71.8 ±8.3
Disease classification (Q1-7)	94.8 ±11.4	89.4 ±13.8	96.0 ±9.5	52.4 ±23.0	70.9 ±20.4	68.8 ±17.4
Intensity calculation (Q8-10)	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	6.4 ±16.7	22.6 ±30.2	9.8 ±23.4
Safety guidelines (Q11-13)	100.0 ±0.0	99.6 ±3.5	99.3 ±4.9	86.7 ±21.1	94.4 ±14.6	94.9 ±12.3
Exercise prescription (Q14-18)	100.0 ±0.0	99.7 ±2.3	100.0 ±0.0	96.8 ±8.3	99.4 ±3.4	99.6 ±3.0

Domain-specific Performance. Domain-specific scores varied across the four clinical evaluation categories. Frontier models achieved 100.0±0.0% on exercise intensity calculation, 99.3 to 100.0% on safety guidelines, and 99.7 to 100.0% on exercise prescription quality. Disease classification ranged from 89.4% (Claude 4.5 Opus) to 96.0% (Gemini 3.0 Pro). Open-weight models scored 96.8 to 99.6% on exercise prescription, 86.7 to 94.9% on safety guidelines, 6.4 to 22.6% on intensity calculation, and 52.4 to 70.9% on disease classification.

Knowledge-Application Gap Analysis. The knowledge-application gap between exam accuracy and clinical vignette performance is shown in Figure 3. Frontier models exhibited gaps ranging from +0.5 to +15.1 percentage points across domains. Open-weight models showed negative gaps in intensity calculation (−33.2 to −52.7 points), positive gaps in safety guidelines (+32.4 to +46.6 points) and exercise prescription (+37.1 to +56.7 points).

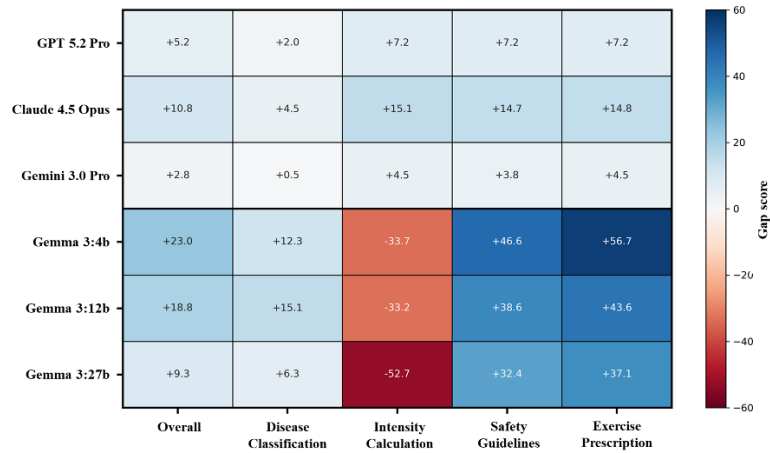


Figure 3: Knowledge-application gap between exam and clinical vignette performance

4. Discussion

We developed comprehensive Korean benchmark evaluating large language models on both theoretical knowledge and clinical exercise prescription for metabolic syndrome and sarcopenia. Assessment of six models across 500 certification exam questions and 120 KNHANES-derived clinical vignettes revealed systematic knowledge-application gaps that varied by task type, establishing clear boundaries for safe clinical deployment.

Non-linear multimodal scaling challenges architectural assumptions. Smaller Gemma models (3:4b, 3:12b) exhibited smaller modality gaps than the larger 3:27b variant, contradicting conventional scaling paradigms. This pattern suggests that multimodal performance may not scale monotonically with parameter count, potentially reflecting differences in vision–language alignment and/or floor effects at low baseline accuracy. One possible explanation is that larger language models may amplify vision encoder errors through more confident but incorrect interpretations, though this hypothesis requires further investigation with controlled experiments. These findings extend observations that medical vision-language models struggle with cross-modal semantic alignment despite architectural advances [18, 37] and suggest that improving vision encoders may yield greater returns than scaling language model parameters for multimodal medical reasoning.

Automated evaluation reliability depends critically on objective verifiability. Our validated LLM-as-a-judge framework achieved overall substantial reliability (ICC 0.761, kappa 0.760), but item-level analysis revealed a threshold effect: objective criteria with algorithmic ground truth (FITT completeness, quantitative calculations) achieved 89 to 100% agreement, whereas two items showed reduced consistency: metabolic syndrome diagnosis requiring integration of five clinical criteria (63.3%) and internal consistency verification of heart rate calculations (58.3%). This pattern aligns with recent frameworks emphasizing that LLM-as-a-judge reliability degrades for tasks lacking clear verification procedures [38, 39]. The findings establish empirical boundaries for scalable automated evaluation in medical contexts and indicate that hybrid human-AI strategies remain necessary for safety-critical domains.

Asymmetric knowledge-application dissociation reveals distinct reasoning failure modes. Frontier models demonstrated uniformly positive knowledge-application gaps, with Gemini 3.0 Pro showing the smallest discrepancy and Claude 4.5 Opus the largest, indicating consistent translation from exam knowledge to clinical application albeit with varying efficiency. Open-weight models exhibited striking bidirectional dissociation: catastrophic failures in intensity calculations (large negative gaps) coexisted with strong performance in safety guidelines and exercise prescription quality (large positive gaps). This pattern reveals that passive knowledge recognition does not guarantee active procedural execution for multi-step numerical tasks, whereas structured output formats enable competent responses even when underlying knowledge is limited. These findings challenge the assumption that exam performance uniformly predicts clinical capability and extend recent evidence that knowledge-based benchmarks inadequately assess practice-based competencies [40, 41].

Limitations. Our benchmark focuses on metabolic syndrome and sarcopenia. Generalizability to other clinical domains, age groups, and international contexts remains untested. The LLM-as-a-judge validation comprised only 100 samples, potentially underestimating edge case failures. Multimodal questions constituted only 13.4% of exam materials, potentially underrepresenting real-world clinical scenarios where visual data interpretation of ECG tracings and body composition scans is essential for routine exercise prescription. Training data contamination cannot be definitively excluded for proprietary models with undisclosed training corpora, though 2023 performance decrements across all models suggest genuine difficulty fluctuations rather than memorization artifacts. Our evaluations represent January 2026 model versions; subsequent updates may substantially alter performance characteristics. Additionally, our reliance on a single expert rater for validation may introduce individual bias, and the absence of formal statistical comparisons limits the strength of our conclusions regarding model differences. The benchmark's focus on Korean clinical guidelines may also limit direct applicability to other healthcare systems.

Implications for deployment and policy. Our findings suggest that exam-based certification alone may be insufficient to assess clinical deployment readiness. Practice-based evaluations incorporating realistic patient scenarios, multimodal data integration, and quantitative reasoning tasks could provide complementary evidence for LLM capabilities in clinical contexts. Open-weight models showed substantial limitations in autonomous intensity calculations, and even frontier models may exhibit failures under distribution shift supporting the need for human oversight in safety-critical numerical tasks. Establishing standardized reporting practices for LLM benchmark studies, such as providing detailed evaluation protocols and validation metrics, would enable meaningful cross-study comparisons and support evidence-based decisions regarding clinical implementation.

Conclusion. We developed the first Korean benchmark evaluating LLMs on both theoretical knowledge and clinical exercise prescription, revealing substantial knowledge-application gaps with distinct failure modes across task types. Open-weight models exhibited catastrophic failures in quantitative reasoning despite moderate exam performance, while frontier models achieved 95 to 98% accuracy but showed persistent multimodal limitations. The validated LLM-as-a-judge framework enables scalable assessment for objective criteria but requires human oversight for subjective clinical judgments. Comprehensive practice-based evaluations incorporating realistic scenarios, multimodal integration, and safety-critical decision-making are essential prerequisites for medical LLM deployment.

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AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: Yes

Justification: The abstract and introduction clearly state the development of a Korean benchmark, the evaluation of six LLMs, and the key finding regarding the knowledge-application gap, which are supported by the results in Section 3.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: Yes

Justification: The abstract and introduction clearly state the development of a Korean benchmark, the evaluation of six LLMs, and the key finding regarding the knowledge-application gap, which are supported by the results in Section 3.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: N/A

Justification: This paper presents an empirical benchmark evaluation of large language models and does not propose new theoretical theorems or proofs.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: Yes

Justification: The paper provides detailed prompt templates (Supplementary Figures 3-5), specific model versions (Section 2.2.1), and public data sources (KSPO, KNHANES) necessary for reproduction.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: No

Justification: While the raw data sources (KSPO, KNHANES) are publicly available via the URLs provided in Section 2.1, the specific curated JSON files and evaluation scripts developed for this study are not explicitly linked in the current manuscript.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: Yes

Justification: Section 2.2 specifies the use of default sampling parameters for all models to simulate typical deployment and details the prompt engineering strategies used.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: Yes

Justification: Results are reported as mean $\pm$ SD across three independent runs (Section 2.3.1), and inter-rater reliability is assessed using appropriate statistical measures (ICC, Cohen's

kappa, etc.) with p-values (Section 3.2).

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: No

Justification: Section 2.2.1 mentions utilizing API endpoints for frontier models and local deployment (Ollama) for open-weight models, but specific hardware specifications (e.g., GPU type, RAM) for the local inferences are not detailed.

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://nips.cc/public/EthicsGuidelines>?

Answer: Yes

Justification: The study relies on publicly available, anonymized data (KNHANES) collected under strict ethical guidelines and IRB approval, adhering to the Declaration of Helsinki (Section 2.1.2).

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: Yes

Justification: Section 4 discusses the implications of using LLMs for safety-critical exercise prescriptions, highlighting the risk of numerical errors endangering vulnerable patients and the necessity of human oversight.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: N/A

Justification: The paper evaluates existing public models and does not release new models or datasets that pose high risks for misuse.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: Yes

Justification: The paper properly cites the original sources for the KSPO exam data and KNHANES data, as well as the technical reports for all LLMs used (References).

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: No

Justification: The paper describes the creation of the benchmark dataset (Section 2.1) and provides examples, but a direct download link or repository for the new benchmark asset is not included in the text.

14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: N/A

Justification: The study is a secondary analysis of existing public data and does not involve crowdsourcing or direct human subject experimentation.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: Yes

Justification: Section 2.1.2 explicitly states that the original KNHANES data collection was approved by the KDCA IRB and provides the specific approval numbers.

Supplemental Material

Supplementary Table 1.

Distribution of multimodal and text-only questions (2020–2024)

Supplementary Figure 1.

Representative multimodal question and JSON structure

Supplementary Figure 2.

Representative example of a clinical vignette generated from KNHANES

Supplementary Figure 3.

Prompt design for the certification examination task

Supplementary Figure 4.

System prompt and output constraints for exercise prescription generation

Supplementary Table 2.

Detailed scoring rubric (checklist) for clinical exercise

Supplementary Figure 5.

System prompt for the automated evaluation pipeline (LLM-as-a-Judge)

Supplementary Table 1: Distribution of multimodal and text-only questions (2020–2024)

Year	Subject Domain	Multimodal	Text-only	Total
2020	Exercise physiology	4	16	20
	Physical fitness assessment	3	17	20
	Exercise prescription	0	20	20
	Exercise stress testing	2	18	20
	Clinical exercise physiology	0	20	20
2021	Exercise physiology	3	17	20
	Physical fitness assessment	5	15	20
	Exercise prescription	0	20	20
	Exercise stress testing	2	18	20
	Clinical exercise physiology	0	20	20
2022	Exercise physiology	2	18	20
	Physical fitness assessment	4	16	20
	Exercise prescription	0	20	20
	Exercise stress testing	3	17	20
	Clinical exercise physiology	2	18	20
2023	Exercise physiology	4	16	20
	Physical fitness assessment	7	13	20
	Exercise prescription	1	19	20
	Exercise stress testing	6	14	20
	Clinical exercise physiology	1	19	20
2024	Exercise physiology	2	18	20
	Physical fitness assessment	8	12	20
	Exercise prescription	1	19	20
	Exercise stress testing	6	14	20
	Clinical exercise physiology	1	19	20
Total	All Domains (2020–2024)	67	433	500



```
{
  "question_number": "xx",
  "year": "20xx",
  "subject": "Exercise Physiology",
  "question_text": "Which of the following correctly lists all the
characteristics appearing in Section A of the resting ECG shown in
the <Box>?",
  "context": "a. Atrial pressure decreases and then increases.\\n b.
Ventricular pressure increases and then decreases.\\n c. Ventricular
volume continuously decreases.\\n d. Aortic pressure increases and
then decreases until the aortic valve closes.",
  "choices": "1) a\\n2) a, b\\n3) a, b, c\\n4) a, b, c, d",
  "img_url": "images/2020_exercise_physiology_10.png"
}
```

Supplementary Figure 1: Representative multimodal question and JSON structure

[Clinical Vignette Content]

1. Patient Profile

Age/Sex: 34-year-old Male

2. Anthropometrics and Body Composition

Height: 179.4 cm

Weight: 78.2 kg

Waist Circumference: 90.0 cm

Body Fat Mass: 18.6 kg

Appendicular Skeletal Muscle Mass (ASM): 25.2 kg

3. Functional Capacity

Grip Strength: 43.8 kg

4. Cardiovascular and Metabolic Parameters

Resting Heart Rate: 80 bpm

Blood Pressure: 123/80 mmHg

Fasting Glucose: 104 mg/dL

Triglycerides: 132 mg/dL

HDL Cholesterol: 44 mg/dL

5. Medication History

Current Medication: Antidiabetic medication

Supplementary Figure 2: Representative example of a clinical vignette generated from KNHANES

[Prompt]

“You are an ACSM Clinical Exercise Physiologist (CEP).

Read the following question and select the correct answer from options 1), 2), 3), and 4).

Please respond with the option number only.

Question: {row['question_text']}”

Supplementary Figure 3: Prompt design for the certification examination task

[Prompt] # Role You are a Chief Clinical Exercise Physiologist at a university hospital sports medicine center. Analyze the provided Patient Data (Vignette) and generate a personalized exercise prescription based on ACSM (American College of Sports Medicine) guidelines. # Task Rules (Strict Adherence) 1. No JSON Output: Write in Markdown format that is easy for evaluators to read. 2. Structure Compliance: Follow the section and header structure of the [Output Template] below exactly, without changing a single word. 3. Explicit Calculation: When setting Target Heart Rate (Target HR), you must include the Karvonen formula and specific values (bpm). 4. Conciseness: Omit the introduction (greetings) and conclusion (closing remarks); output only the main content. # Output Template ## 1. Risk Assessment - Key Risk Factors: (List risk factors based on the patient's test values and medication history. e.g., Abdominal obesity, Hypertension, etc.) - Metabolic Syndrome Status: (Yes/No) - Sarcopenia Status: (Yes/No, based on AWGS 2019 criteria: Low muscle strength + Low muscle mass) ## 2. Target Heart Rate Configuration *(Apply Karvonen Formula: $HRR = Max\ HR - Resting\ HR$)* *(Max HR = $220 - Age$)* - Estimated Max HR: XX bpm - Resting HR: XX bpm - Heart Rate Reserve (HRR): XX bpm - Light Intensity (30-39%): XX ~ XX bpm - Moderate Intensity (40-59%): XX ~ XX bpm - Vigorous Intensity (60-89%): XX ~ XX bpm ## 3. Safety Precautions - Elderly/Sarcopenia: (Guidelines related to fall prevention, balance training, etc.) - Medication-Related: (Specific precautions based on medications such as insulin/antidiabetics, antihypertensives, lipid-lowering drugs, etc.) - Disease-Specific Contraindications: (Contraindicated movements and breathing techniques based on underlying conditions like hypertension) ## 4. FITT Exercise Prescription Component Aerobic Exercise Resistance Exercise :--- :--- :--- Frequency (F) X times/week X times/week Intensity (I) (Specific HR range in bpm or RPE) (% 1RM or RPE) Time (T) X min/day (or total weekly time) Sets/Reps Type (T) (Specific modality recommendation) (Specific modality recommendation) # Patient Data {vignette}

Supplementary Figure 4. System prompt and output constraints for exercise prescription generation

Supplementary Table 2. Detailed scoring rubric (checklist) for clinical exercise

Category	Item	Evaluation Criteria (Question)	Scoring
Disease Classification	Q1	Abdominal Obesity: Did the model correctly identify the presence/absence based on waist circumference criteria?	0 / 1
	Q2	Hypertension: Did the model correctly identify the presence/absence based on BP values or medication history?	0 / 1
	Q3	Hyperglycemia: Did the model correctly identify the presence/absence based on fasting glucose or medication history?	0 / 1
	Q4	Hypertriglyceridemia: Did the model correctly identify the presence/absence based on triglyceride levels or lipid-lowering medication?	0 / 1
	Q5	Low HDL-Cholesterol: Did the model correctly identify the presence/absence based on HDL levels or lipid-lowering medication?	0 / 1
	Q6	Metabolic Syndrome Diagnosis: Did the model correctly diagnose "Yes" if ≥ 3 criteria were met, or "No" otherwise?	0 / 1
	Q7	Sarcopenia Diagnosis: Did the model correctly diagnose based on AWGS 2019 criteria (Low muscle mass + Low grip strength)?	0 / 1
Exercise Intensity	Q8	Light Intensity Accuracy: Does the calculated heart rate range (30–39% HRR) match the ground truth within ± 5 bpm?	0 / 1
	Q9	Moderate Intensity Accuracy: Does the calculated heart rate range (40–59% HRR) match the ground truth within ± 5 bpm?	0 / 1
	Q10	Vigorous Intensity Accuracy: Does the calculated heart rate range (60–89% HRR) match the ground truth within ± 5 bpm?	0 / 1
Safety Guidelines	Q11	Elderly/Fall Prevention: Did the model mention fall prevention or balance training for elderly/sarcopenic patients?	0 / 1
	Q12	Hypoglycemia Prevention: Did the model mention precautions for hypoglycemia or snack intake for patients on insulin/antidiabetics?	0 / 1
	Q13	Hypertension Safety: Did the model mention precautions such as avoiding the Valsalva maneuver or orthostatic hypotension for patients on antihypertensives?	0 / 1
Exercise Prescription	Q14	Structural Completeness: Are all FITT fields (Frequency, Intensity, Time, Type) filled for both aerobic and resistance exercises?	0 / 1
	Q15	Aerobic Guideline Adherence: Does the prescription meet frequency (3–5+ days/week) and duration (20–60 min) guidelines?	0 / 1
	Q16	Resistance Guideline Adherence: Does the prescription meet frequency (2–3 days/week) and specify appropriate intensity/load?	0 / 1
	Q17	Internal Consistency: Do the HR ranges listed in the FITT table match the calculated ranges in the "Target Heart Rate" section?	0 / 1
	Q18	Appropriateness of Modality: Did the model recommend suitable exercise types while avoiding contraindicated movements for the patient's condition?	0 / 1

[Prompt]

Role

You are a **Chief Evaluator** with profound expertise in **ACSM Guidelines** and **Clinical Exercise Physiology**. Your mission is to review the exercise prescription generated by a candidate AI model and strictly grade its **Medical Accuracy** and **Safety** based on the patient's actual data (Vignette).

Evaluation Process (Adaptive Reasoning)

The evaluation must proceed through the following **two steps**:

Step 1. Establish Ground Truth

Analyze the provided Patient Data and Ground Truth to confirm the correct answers. Prioritize the provided Ground Truth, but additionally verify the following logic:

1. Identification of 5 Metabolic Syndrome Factors (NCEP-ATP III Korean Criteria + Medication Consideration):

***Medication Mapping Rules:**

- Intake of "Antihypertensives" → Consider as **Hypertension**.
- Intake of "Antidiabetic agents" or "Insulin" → Consider as **Hyperglycemia**.
- Intake of "Lipid-lowering drugs" → Consider as Dyslipidemia (High TG/Low HDL).

***Diagnostic Cutoffs:**

- Abdominal Obesity: Male ≥ 90 cm, Female ≥ 85 cm
- Hypertension: $\geq 130/85$ mmHg
- Hyperglycemia: ≥ 100 mg/dL
- Hypertriglyceridemia: ≥ 150 mg/dL
- Low HDL-Cholesterol: Male < 40 mg/dL, Female < 50 mg/dL

2. Sarcopenia Determination (AWGS 2019):

- Classify as **True** only if both Low Muscle Strength (Grip: M < 28 kg, F < 18 kg) AND Low Muscle Mass (SMI: M < 7.0 kg/m², F < 5.7 kg/m²) are present.

3. Target Heart Rate Calculation (Karvonen Formula):

- $HRR = (220 - \text{Age}) - \text{Resting HR}$
- **Light Intensity:** $(HRR \times 0.30 \sim 0.39) + \text{Resting HR}$
- **Moderate Intensity:** $(HRR \times 0.40 \sim 0.59) + \text{Resting HR}$
- **Vigorous Intensity:** $(HRR \times 0.60 \sim 0.89) + \text{Resting HR}$
- *(Round to the nearest whole number)*

Step 2. Precise Boolean Scoring

Compare your calculated 'Step 1 Ground Truth' with the 'Candidate AI Response' and score each item as Correct (1) or Incorrect (0).

Scoring Rubrics (Checklist)

(Refer to the detailed criteria provided in the context)

Input Data

Patient Data (Vignette)

{vignette}

Ground Truth (Verified Data)

The following is pre-validated ground truth data. Use this as the reference for grading.
{ground_truth}

Exercise Prescription to Evaluate (Generated by AI)

{prescription}

Supplementary Figure 5. System prompt for the automated evaluation pipeline (LLM-as-a-Judge)